

1. OHM'S LAW & POWER WHEEL

To Find	Given I & R	Given V & R	Given V & I
Voltage (V)	$V = I \times R$	—	—
Current (I)	—	$I = V \div R$	—
Resistance (R)	—	—	$R = V \div I$

V = volts, I = amps, R = ohms. Cover the unknown quantity, the wheel gives you the operation.

POWER FORM	FORMULA
Given I & R	$P = I^2 \times R$
Given V & R	$P = V^2 \div R$
Given V & I	$P = V \times I$

P = watts. Same cover-the-unknown logic applies to the power wheel.

2. POWER FORMULAS — SINGLE & THREE-PHASE

CIRCUIT	FORMULA	NOTES
DC / Single-phase (no PF given)	$P = V \times I$	Watts, resistive load
Single-phase AC (with power factor)	$P = V \times I \times PF$	PF from nameplate or given in problem
Three-phase (with power factor)	$P = V \times I \times 1.732 \times PF$	$1.732 = \sqrt{3}$, always shows up in 3-phase calcs
Three-phase current, given kVA	$I = kVA \times 1000 \div (V \times 1.732)$	Common motor/service sizing question

3. VOLTAGE DROP — FORMULA & WORKED EXAMPLE

CIRCUIT	FORMULA
Single-phase	$VD = (2 \times K \times I \times D) \div CM$
Three-phase	$VD = (1.732 \times K \times I \times D) \div CM$

K = resistivity (≈ 12.9 copper, ≈ 21.2 aluminum) · I = amps · D = one-way distance (ft) · CM = circular mils of conductor.

Problem: 120V single-phase circuit, 20A load, 100 ft one-way, 12 AWG copper ($CM = 6,530$).

Formula: $VD = (2 \times 12.9 \times 20 \times 100) \div 6,530$

Calc: $VD = 51,600 \div 6,530 = 7.90$ volts

Percent drop: $7.90 \div 120 = 6.58\%$

Result: Exceeds the common 3% branch-circuit design target — upsize the conductor or shorten the run.

4. CONDUCTOR AMPACITY

Ampacity values depend on conductor material, insulation type, and ambient/adjustment factors — **do not memorize a single number**. Always verify the actual allowable ampacity in **NEC Table 310.16** (and apply correction factors from 310.15(B) as needed) for the specific conductor and conditions in the question.

5. GROUNDING VS. BONDING

Grounding — connecting electrical systems/equipment to the earth (or a conductive body that serves in place of the earth) to control voltage during fault conditions and limit overvoltage.

Bonding — connecting metal parts together to ensure electrical continuity and a low-impedance fault-current path, so overcurrent devices clear faults quickly.

6. EXAM-DAY CHECKLIST

- Bring the NEC code book edition your state board requires for open-book exams — **this varies by state**; confirm with your state licensing board before test day.
- Check whether your book may be tabbed/highlighted — rules vary by state and testing center.
- Confirm allowed calculator model/type — some states require a specific approved list, others restrict to non-programmable calculators; **varies by state**.
- Bring required photo ID and any confirmation/registration paperwork the testing center specifies.
- Arrive early — late arrival policies vary by testing center and may forfeit your exam slot.
- Know your time limit and question count in advance so you can pace yourself.